

**INTEGRAL CALCULUS**

**SINGLE CORRECT TYPE QUESTIONS**

1. If  $\theta_1$  and  $\theta_2$  be respectively the smallest and the largest values of  $\theta$  in  $(0, 2\pi) - \{\pi\}$  which satisfy the equation,  $2\cot^2\theta - \frac{5}{\sin\theta} + 4 = 0$  then  $\int_{\theta_1}^{\theta_2} \cos^2 3\theta d\theta$  is equal to :

$(0, 2\pi) - \{\pi\}$  में समीकरण  $2\cot^2\theta - \frac{5}{\sin\theta} + 4 = 0$  को संतुष्ट करने वाले  $\theta$  के न्यूनतम तथा अधिकतम मान क्रमशः  $\theta_1$  तथा  $\theta_2$  हैं, तो  $\int_{\theta_1}^{\theta_2} \cos^2 3\theta d\theta$  बराबर है :

$\theta_2$  हैं, तो  $\int_{\theta_1}^{\theta_2} \cos^2 3\theta d\theta$  बराबर है :

- (A)  $\frac{\pi}{3}$                       (B)  $\frac{2\pi}{3}$                       (C)  $\frac{\pi}{3} + \frac{1}{6}$                       (D)  $\frac{\pi}{9}$

Ans. A

**Sol.**  $2\cot^2\theta - 5\operatorname{cosec}\theta + 4 = 0$   
 $2(\operatorname{cosec}^2\theta - 1) - 5\operatorname{cosec}\theta + 4 = 0$   
 $2\operatorname{cosec}^2\theta - 5\operatorname{cosec}\theta + 2 = 0$   
 $(2\operatorname{cosec}\theta - 1)(\operatorname{cosec}\theta - 2) = 0$   
 $\operatorname{cosec}\theta = 2$   
 $\Rightarrow \theta = \frac{\pi}{6}, \frac{5\pi}{6}$

$$I = \int_{\pi/6}^{5\pi/6} \cos^2(3\theta) d\theta$$

$$= \int_{\pi/6}^{5\pi/6} \frac{1 + \cos 6\theta}{2} d\theta$$

$$= \pi/3$$

2. If  $y = \left(\frac{2}{\pi}x - 1\right)\operatorname{cosec}x$  is the solution of the differential equation,  $\frac{dy}{dx} + p(x)y = \frac{2}{\pi}\operatorname{cosec}x, 0 < x < \frac{\pi}{2}$ , then the function  $p(x)$  is equal to :

यदि अवकल समीकरण  $\frac{dy}{dx} + p(x)y = \frac{2}{\pi}\operatorname{cosec}x, 0 < x < \frac{\pi}{2}$ , का हल  $y = \left(\frac{2}{\pi}x - 1\right)\operatorname{cosec}x$  है तो फलन  $p(x)$  बराबर है:

- (A)  $\tan x$                       (B)  $\sec x$                       (C)  $\cot x$                       (D)  $\operatorname{cosec} x$

Ans. C

Sol.  $y = \left( \frac{2x}{\pi} - 1 \right) \operatorname{cosec} x$

$$\frac{dy}{dx} = \left( \frac{2x}{\pi} - 1 \right) (-\operatorname{cosec} x \cot x) + \operatorname{cosec} x \left( \frac{2}{\pi} \right)$$

$$\frac{dy}{dx} = \left( \frac{2x}{\pi} - 1 \right) (\operatorname{cosec} x)(-\cot x) + 2 \frac{\operatorname{cosec} x}{\pi}$$

$$\frac{dy}{dx} + y \cot x = \frac{2 \operatorname{cosec} x}{\pi}$$

$$\frac{dy}{dx} + P(x)y = \frac{2}{\pi} \operatorname{cosec} x$$

$$p(x) = \cot x$$

3. The value of the integral,  $\int_1^3 [x^2 - 2x - 2] dx$ , where  $[x]$  denotes the greatest integer less than or equal to  $x$ , is:

समाकलन  $\int_1^3 [x^2 - 2x - 2] dx$  का मान, जबकि  $[x]$  महत्तम पूर्णांक  $\leq x$  है, है :

(A)  $-5$  (B)  $-\sqrt{2} - \sqrt{3} + 1$  (C)  $-\sqrt{2} - \sqrt{3} - 1$  (D)  $-4$

Ans. C

Sol.  $I = \int_1^3 [x^2 - 2x - 2] dx = \int_1^3 [(x-1)^2 - 3] dx$

Put  $x - 1 = t$

$$I = \int_0^2 [t^2 - 3] dx = \int_0^2 [t^2] dx - 3 \int_0^2 dx$$

$$= \int_0^1 0 dt + \int_1^{\sqrt{2}} 1 dt + \int_{\sqrt{2}}^{\sqrt{3}} 2 dt + \int_{\sqrt{3}}^2 3 dt - 6$$

$$= -\sqrt{2} - \sqrt{3} - 1$$

4. The integral  $\int_{\pi/6}^{\pi/4} \frac{dx}{\sin 2x (\tan^5 x + \cot^5 x)}$  equals :

समाकल  $\int_{\pi/6}^{\pi/4} \frac{dx}{\sin 2x (\tan^5 x + \cot^5 x)}$  बराबर है -

- (A)  $\frac{\pi}{40}$  (B)  $\frac{1}{20} \tan^{-1} \left( \frac{1}{9\sqrt{3}} \right)$  (C)  $\frac{1}{10} \left( \frac{\pi}{4} - \tan^{-1} \left( \frac{1}{9\sqrt{3}} \right) \right)$  (D)  $\frac{1}{5} \left( \frac{\pi}{4} - \tan^{-1} \left( \frac{1}{3\sqrt{3}} \right) \right)$

Ans. C

Sol.  $\int_{\pi/6}^{\pi/4} \frac{dx}{\sin 2x (\tan^5 x + \cot^5 x)}$

$$\frac{1}{2} \int_{\pi/6}^{\pi/4} \frac{dx}{\sin x \cos x \left( \frac{\tan^{10} x + 1}{\tan^5 x} \right)}$$

$$\frac{1}{2} \int_{\pi/6}^{\pi/4} \frac{\tan^5 x dx}{\sin x \cos x (\tan^{10} x + 1)} = \frac{1}{2} \int_{\pi/6}^{\pi/4} \frac{\tan^4 x \sec^2 x dx}{1 + \tan^{10} x}$$

$$\tan x = t$$

$$\Rightarrow \sec^2 x dx = dt$$

$$= \frac{1}{2} \int_{1/3}^1 \frac{t^4 dt}{1 + t^{10}} \quad t^5 = 4$$

$$5t^4 dt = dU$$

$$t^4 dt = \frac{dU}{5}$$

$$= \frac{1}{10} \int_{1/3}^1 \frac{dU}{1 + U^2} = \frac{1}{10} \tan^{-1}(U)$$

5. If  $\int_0^{\pi/2} \frac{\cot x}{\cot x + \operatorname{cosec} x} dx = m(\pi + n)$ , then  $m \cdot n$  is equal to :

यदि  $\int_0^{\pi/2} \frac{\cot x}{\cot x + \operatorname{cosec} x} dx = m(\pi + n)$ , तो  $m \cdot n$  बराबर है :

- (A)  $\frac{1}{2}$  (B) 1 (C) -1 (D)  $-\frac{1}{2}$

Ans. C

sol.  $\int_0^{\pi/2} \frac{\cot x dx}{\cot x + \operatorname{cosec} x}$

$$\int_0^{\pi/2} \frac{\cos x dx}{1 + \cos x} = \int_0^{\pi/2} \left( 1 - \frac{1}{1 + \cos x} \right) dx$$

$$= [x]_0^{\pi/2} - \int_0^{\pi/2} \frac{1}{2 \cos^2 \frac{x}{2}} dx$$

$$= \frac{\pi}{2} - \frac{1}{2} \int_0^{\pi/2} \sec^2 \frac{x}{2} dx$$

$$= \frac{\pi}{2} - \left[ \tan \frac{x}{2} \right]_0^{\pi/2}$$

$$= \frac{\pi}{2} - [1] = \left( \frac{\pi}{2} - 1 \right)$$

$$m = \frac{1}{2}, n = -2$$

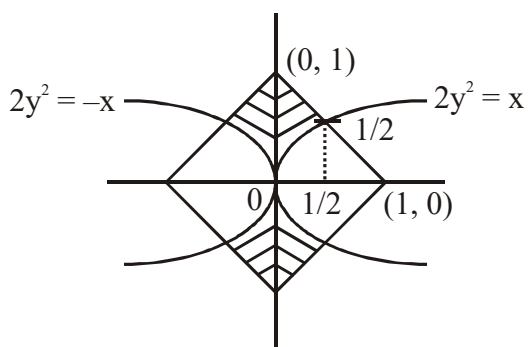
$$\Rightarrow mn = -1$$

6. The area (in sq. units) of the region  $A = \{(x, y) : |x| + |y| \leq 1, 2y^2 \geq |x|\}$  is :

क्षेत्र  $A = \{(x, y) : |x| + |y| \leq 1, 2y^2 \geq |x|\}$  का क्षेत्रफल ( वर्ग इकाइयों में ) है :

- (A)  $\frac{7}{6}$  (B)  $\frac{1}{6}$  (C)  $\frac{1}{3}$  (D)  $\frac{5}{6}$

Ans. D



Sol.

$$A = 4 \left( \frac{1}{2} \times 1 \times 1 - \int_0^{\frac{1}{2}} \sqrt{\frac{x}{2}} dx - \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \right)$$

$$A = \frac{5}{6}$$

7. If  $\int \frac{d\theta}{\cos^2 \theta (\tan 2\theta + \sec 2\theta)} =$

$\lambda \tan \theta + 2 \log_e |f(\theta)| + C$  where  $C$  is a constant of integration, then the ordered pair  $(\lambda, f(\theta))$  is equal to :

यदि  $\int \frac{d\theta}{\cos^2 \theta (\tan 2\theta + \sec 2\theta)} =$

$\lambda \tan \theta + 2 \log_e |f(\theta)| + C$  है, जहाँ  $C$  एक समाकलन अचर है, तो क्रमित युग्म  $(\lambda, f(\theta))$  बराबर है :

(A)  $(1, 1 - \tan \theta)$  (B)  $(1, 1 + \tan \theta)$  (C)  $(-1, 1 + \tan \theta)$  (D)  $(-1, 1 - \tan \theta)$

Ans. C

**Sol.**  $\int \frac{d\theta}{\cos^2 \theta (\tan 2\theta + \sec 2\theta)} = \lambda \tan \theta + 2 \ln |f(\theta)| + c$

$$\int \frac{d\theta}{\cos^2 \theta \left( \frac{1 + \sin 2\theta}{\cos 2\theta} \right)} = \int \frac{\cos 2\theta d\theta}{\cos^2 \theta (1 + \sin 2\theta)}$$

$$= \int \frac{(\cos \theta - \sin \theta)(\cos \theta + \sin \theta)}{\cos^2 \theta (\cos \theta + \sin \theta)^2} d\theta$$

$$= \int \sec^2 \theta \left( \frac{1 - \tan \theta}{1 + \tan \theta} \right) d\theta$$

$$\tan \theta = t \quad \Rightarrow \quad \sec^2 \theta d\theta = dt$$

$$\int \left( \frac{1-t}{1+t} \right) dt$$

$$1+t = u$$

$$dt = du$$

$$\int \frac{1-(u-1)}{u} du = \int \frac{2-u}{u} du$$

$$2 \int \frac{du}{u} - \int du = 2 \ln |u| - u + C$$

$$= 2 \ln |1+t| - (1+t) + C$$

$$= 2 \ln |1 + \tan \theta| - (1 + \tan \theta) + C$$

$$= 2 \ln |1 + \tan \theta| - \tan \theta + C$$

8. The general solution of the differential equation  $\sqrt{1+x^2+y^2+x^2y^2} + xy \frac{dy}{dx} = 0$  is :

(where C is a constant of integration)

अवकल समीकरण  $\sqrt{1+x^2+y^2+x^2y^2} + xy \frac{dy}{dx} = 0$  का व्यापक हल है :

(जहाँ C एक समाकलन अचर है)

(A)  $\sqrt{1+y^2} + \sqrt{1+x^2} = \frac{1}{2} \log_e \left( \frac{\sqrt{1+x^2}+1}{\sqrt{1+x^2}-1} \right) + C$

(B)  $\sqrt{1+y^2} - \sqrt{1+x^2} = \frac{1}{2} \log_e \left( \frac{\sqrt{1+x^2}-1}{\sqrt{1+x^2}+1} \right) + C$

(C)  $\sqrt{1+y^2} + \sqrt{1+x^2} = \frac{1}{2} \log_e \left( \frac{\sqrt{1+x^2}-1}{\sqrt{1+x^2}+1} \right) + C$

(D)  $\sqrt{1+y^2} - \sqrt{1+x^2} = \frac{1}{2} \log_e \left( \frac{\sqrt{1+x^2}+1}{\sqrt{1+x^2}-1} \right) + C$

Ans. A

Sol.  $\sqrt{(1+x^2)(1+y^2)} = -xy \frac{dy}{dx}$

$$\int \frac{\sqrt{1+x^2}}{x} dx = \int \frac{-y}{\sqrt{1+y^2}} dy$$

$$\int \frac{\sqrt{1+x^2}}{x^2} (x dx) = \int \frac{-y}{\sqrt{1+y^2}} dy$$

$$\begin{aligned} 1+x^2 &= t^2 & 1+y^2 &= k^2 \\ x dx &= t dt & y dy &= k dk \end{aligned}$$

$$\int \frac{t^2}{t^2-1} dt = \int -dk$$

$$\int \left( 1 + \frac{1}{t^2-1} \right) dt = -k + C$$

$$t + \frac{1}{2} \ln \left( \frac{t-1}{t+1} \right) = -k + C$$

$$\sqrt{1+x^2} + \frac{1}{2} \ln \left( \frac{\sqrt{1+x^2}-1}{\sqrt{1+x^2}+1} \right) = -\sqrt{1+y^2} + C$$

$$\sqrt{1+x^2} + \sqrt{1+y^2} = \frac{1}{2} \ln \left( \frac{\sqrt{1+x^2}+1}{\sqrt{1+x^2}-1} \right) + C$$

9. If  $\int \frac{dx}{(x^2-2x+10)^2} = A \left( \tan^{-1} \left( \frac{x-1}{3} \right) + \frac{f(x)}{x^2-2x+10} \right) + C$

where C is a constant of integration, then :

(A)  $A = \frac{1}{81}$  and  $f(x) = 3(x-1)$

(B)  $A = \frac{1}{54}$  and  $f(x) = 3(x-1)$

(C)  $A = \frac{1}{27}$  and  $f(x) = 9(x-1)$

(D)  $A = \frac{1}{54}$  and  $f(x) = 9(x-1)^2$

यदि  $\int \frac{dx}{(x^2-2x+10)^2} = A \left( \tan^{-1} \left( \frac{x-1}{3} \right) + \frac{f(x)}{x^2-2x+10} \right) + C$

जहाँ C एक समाकलन अचर है, तो :

(A)  $A = \frac{1}{81}$  तथा  $f(x) = 3(x-1)$

(B)  $A = \frac{1}{54}$  तथा  $f(x) = 3(x-1)$

(C)  $A = \frac{1}{27}$  तथा  $f(x) = 9(x-1)$

(D)  $A = \frac{1}{54}$  तथा  $f(x) = 9(x-1)^2$

Ans. B

sol.  $\int \frac{dx}{(x^2-2x+10)^2} = \int \frac{dx}{((x-1)^2+9)^2}$

Let  $(x-1)^2 = 9 \tan^2 \theta$  ....(i)

$\Rightarrow \tan \theta = \frac{x-1}{3}$

On differentiating ....(i)

$2(x-1)dx = 18 \tan \theta \sec^2 \theta d\theta$

$\therefore I = \int \frac{18 \tan \theta \sec^2 \theta d\theta}{2 \times 3 \tan \theta \times 81 \sec^2 \theta}$

$$I = \frac{1}{27} \int \cos^2 \theta d\theta = \frac{1}{27} \times \frac{1}{2} \int (1 + \cos 2\theta) d\theta$$

$$I = \frac{1}{54} \left\{ \theta + \frac{\sin 2\theta}{2} \right\} + c$$

$$I = \frac{1}{54} \left[ \tan^{-1} \left( \frac{x-1}{3} \right) + \frac{1}{2} \times \frac{2 \left( \frac{x-1}{3} \right)}{1 + \left( \frac{x-1}{3} \right)^2} \right] + c$$

$$I = \frac{1}{54} \left[ \tan^{-1} \left( \frac{x-1}{3} \right) + \frac{3(x-1)}{x^2 - 2x + 10} \right] + c$$

$$\text{So } A = \frac{1}{54}$$

$$f(x) = 3(x-1)$$

10. If  $f(a + b + 1 - x) = f(x)$ , for all  $x$ , where  $a$  and  $b$  are fixed positive real numbers,

then  $\frac{1}{a+b} \int_a^b x(f(x) + f(x+1))dx$  is equal to :

यदि सभी  $x$  के लिए,  $f(a + b + 1 - x) = f(x)$  है, जबकि  $a$  तथा  $b$  स्थिर धन वास्तविक संख्याएँ हैं, तो

$\frac{1}{a+b} \int_a^b x(f(x) + f(x+1))dx$  बराबर है :

(A)  $\int_{a-1}^{b-1} f(x+1)dx$     (B)  $\int_{a+1}^{b+1} f(x)dx$     (C)  $\int_{a-1}^{b-1} f(x)dx$     (D)  $\int_{a+1}^{b+1} f(x+1)dx$

Ans. C

**Sol.**  $I = \frac{1}{a+b} \int_a^b x(f(x) + f(x+1))dx$  ....(1)

Using the property  $\int_a^b f(x) dx = \int_a^b f(a+b+x) dx$  ;

$$I = \frac{1}{a+b} \int_a^b (a+b-x)(f(a+b-x) + f(a+b-x+1)) dx$$
 ....(2)

But given that  $f(a + b + 1 - x) = f(x) \Rightarrow (a + b - x) = f(1 + x)$

So Equation (2) becomes

$$I = \frac{1}{a+b} \int_a^b (a+b-x) (f(x+1) + f(x)) dx \quad \dots(3)$$

Adding (1) and (3)  $2I = \int_a^b (f(x+1) + f(x)) dx$

$$\Rightarrow 2I = \int_a^b f(x+1) dx + \int_a^b f(a+b-x) dx$$

$$\Rightarrow 2I = 2 \int_a^b f(x+1) dx$$

$$\Rightarrow I = \int_a^b f(a+b-x) dx$$

Putting  $x+1 = t$ ,  $I = \int_{a-1}^{b-1} f(t) dt = \int_{a-1}^{b-1} f(x) dx$

11. If  $\int (e^{2x} + 2e^x - e^{-x} - 1)e^{(e^x + e^{-x})} dx = g(x)e^{(e^x + e^{-x})} + c$ , where  $c$  is a constant of integration, then  $g(0)$  is equal to:

यदि  $\int (e^{2x} + 2e^x - e^{-x} - 1)e^{(e^x + e^{-x})} dx = g(x)e^{(e^x + e^{-x})} + c$  हैं, जहाँ  $c$  एक समाकलन अचर है, तो  $g(0)$  बराबर है :

- (A) 2 (B)  $e$  (C)  $e^2$  (D) 1

Ans. A

Sol.  $\int (e^{2x} + 2e^x - e^{-x} - 1)e^{(e^x + e^{-x})} dx$

$$= \int (e^x - e^{-x})e^{e^x + e^{-x}} dx + \int (e^{2x} - e^x - 1)e^{e^x + e^{-x}} dx = I_1 + I_2$$

For  $I_1$ ,  $e^x + e^{-x} = t$ ;  $(e^x - e^{-x})dx = dt$

$$I_1 = \int e^t dt = e^t + C = e^{e^x + e^{-x}} + C$$

$$I_2 = \int e^x (e^x - e^{-x} + 1)e^{e^x + e^{-x}} dx$$

$$\int (e^x - e^{-x} + 1)e^{e^x + e^{-x} + x} dx$$

Let  $e^x + e^{-x} + x = u$ ;  $(e^x - e^{-x} + 1)dx = du$

So,  $I_2 = \int e^u du = e^u + C = e^{e^x + e^{-x} + x} + C$

$$I = I_1 + I_2 = e^{e^x + e^{-x}} (e^x + 1) + C$$

$$g(x) = e^x + 1$$

$$g(0) = 2$$

12. Let  $y = y(x)$  be the solution of the differential equation  $\cos x(3 \sin x + \cos x + 3) dy = (1 + y \sin x$

$(3 \sin x + \cos x + 3))dx$ ,  $0 \leq x \leq \frac{\pi}{2}$ ,  $y(0) = 0$ . Then,  $y\left(\frac{\pi}{3}\right)$  is equal to :

माना अवकल समीकरण  $\cos x(3 \sin x + \cos x + 3) dy = (1 + y \sin x (3 \sin x + \cos x + 3))dx$ ,  $0 \leq x \leq \frac{\pi}{2}$ ,  $y(0) = 0$

का हल  $y = y(x)$  है। तो  $y\left(\frac{\pi}{3}\right)$  बराबर है :

(A)  $2 \log_e \left( \frac{2\sqrt{3} + 10}{11} \right)$

(B)  $2 \log_e \left( \frac{\sqrt{3} + 7}{2} \right)$

(C)  $2 \log_e \left( \frac{2\sqrt{3} + 9}{6} \right)$

(D)  $2 \log_e \left( \frac{3\sqrt{3} - 8}{4} \right)$

Ans. A

Sol.  $\cos x (3 \sin x + \cos x + 3) dy = 1 + y \sin x (3 \sin x + \cos x + 3) dx$

$$\frac{dy}{dx} + y(-\tan x) = \frac{1}{\cos x (3 \sin x + \cos x + 3)}$$

$$\int \tan x \, dx$$

$$I.F. = e^{-\int \tan x \, dx}$$

$$I. F. = \cos x$$

$$y (I. F. ) = \int (I.F.) Q \, dx$$

$$y (\cos x) = \int \frac{\cos x \, dx}{\cos x (3 \sin x + \cos x + 3)}$$

$$y(\cos x) = \int \frac{dx}{3 \left( \frac{2 \tan \frac{x}{2}}{1 + \tan^2 \frac{x}{2}} \right) + \left( \frac{1 - \tan^2 \frac{x}{2}}{1 + \tan^2 \frac{x}{2}} \right) + 3}$$

$$y(\cos x) = \int \frac{\sec^2 \frac{x}{2} dx}{6 \tan \frac{x}{2} + \left( 1 - \tan^2 \frac{x}{2} \right) + 3 \left( 1 + \tan^2 \frac{x}{2} \right)}$$

$$y(\cos x) = \int \frac{\sec^2 \frac{x}{2} dx}{2 \tan^2 \frac{x}{2} + 6 \tan^2 \frac{x}{2} + 4}$$

$$y(\cos x) = \frac{1}{2} \int \frac{\sec^2 \frac{x}{2} dx}{\tan^2 \frac{x}{2} + 3 \tan \frac{x}{2} + 2}$$

$$\tan \frac{x}{2} = t$$

$$\frac{1}{2} \sec^2 \frac{x}{2} dx = dt$$

$$y(\cos x) = \int \frac{dt}{t^2 + 3t + 2}$$

$$y(\cos x) = \int \frac{dt}{\left( t + \frac{3}{2} \right)^2 - \left( \frac{1}{2} \right)^2}$$

$$y(\cos x) = \frac{1}{2 \times \frac{1}{2}} \ln \left| \frac{2t + 3 - 1}{2t + 3 + 1} \right| + C$$

$$y(\cos x) = \ln \left| \frac{2t + 2}{2t + 4} \right| + C$$

$$y(\cos x) = \ln \left| \frac{t + 1}{t + 2} \right| + C$$

$$y(\cos x) = \ln \left| \frac{\tan \frac{x}{2} + 1}{\tan \frac{x}{2} + 2} \right| + C$$

$$x = 0 \text{ then } y = 0$$

$$0 = \ln \left| \frac{1}{2} \right| + C$$

$$y(\cos x) = \ln \left| \frac{\tan \frac{x}{2} + 1}{\tan \frac{x}{2} + 2} \right| - \ln \frac{1}{2}$$

$$y \times \frac{1}{2} = \ln \left| \frac{\frac{1}{\sqrt{3}} + 1}{\frac{1}{\sqrt{3}} + 2} \right| - \ln \frac{1}{2}$$

$$\frac{y}{2} = \ln \left| \frac{1 + \sqrt{3}}{1 + 2\sqrt{3}} \right| - \ln \frac{1}{2}$$

$$\frac{y}{2} = \ln \left| \frac{\sqrt{3} + 1}{2\sqrt{3} + 1} \times \frac{2\sqrt{3} - 1}{2\sqrt{3} - 1} \right| - \ln \frac{1}{2}$$

$$\frac{y}{2} = \ln \left| \frac{5 + \sqrt{3}}{11} \right| - \ln \frac{1}{2}$$

$$y = 2 \ln \left| \frac{10 + 2\sqrt{3}}{11} \right|$$

13. The integral  $16 \int_1^2 \frac{dx}{x^3(x^2 + 2)^2}$  is equal to

समाकलन  $16 \int_1^2 \frac{dx}{x^3(x^2 + 2)^2}$  बराबर है

- (A)  $\frac{11}{6} + \log_e 4$       (B)  $\frac{11}{12} + \log_e 4$       (C)  $\frac{11}{6} - \log_e 4$       (D)  $\frac{11}{12} - \log_e 4$

Ans. C

Sol.  $\int_1^2 \frac{dx}{x^3 x^4} \left(1 + \frac{2}{x^2}\right)^2$

Let  $1 + \frac{2}{x^2} = t \Rightarrow \frac{-4}{x^3} dx = dt$

$$I = \frac{-1}{4} \int_3^{\frac{3}{2}} \left(\frac{t-1}{2}\right)^2 \frac{dt}{t^2} = -\frac{1}{16} \int_3^{\frac{3}{2}} \frac{t^2 - 2t + 1}{t^2} dt$$

$$= \frac{-1}{16} \int_3^{\frac{3}{2}} \left(1 - \frac{2}{t} + \frac{1}{t^2}\right) dt$$

$$= \frac{-1}{16} \left( t - 2 \ln |t| - \frac{1}{t} \right)_3^{\frac{3}{2}}$$

$$= \frac{-1}{16} \left( \frac{3}{2} - 2 \ln \frac{3}{2} - \frac{2}{3} \right) - \left( 3 - 2 \ln 3 - \frac{1}{3} \right)$$

$$= \frac{-1}{16} \left( 2 \left( \ln 2 - \ln \frac{3}{2} \right) - \frac{3}{2} - \frac{1}{3} \right)$$

$$= \frac{-1}{16} \left( 2 \ln 3 - \frac{11}{6} \right) = \frac{-1}{16} \left( \ln 4 - \frac{11}{6} \right) = \frac{11 - 6 \ln 4}{16 \times 6}$$

$$\frac{11 - 6 \ln 4}{96}$$

14. The value of  $\int_0^{2\pi} [\sin 2x(1 + \cos 3x)] dx$ , where  $[t]$  denotes the greatest integer function, is :

$\int_0^{2\pi} [\sin 2x(1 + \cos 3x)] dx$  का मान, जहाँ  $[t]$  महत्तम पूर्णांक फलन है, है:

- (A)  $\pi$  (B)  $-\pi$  (C)  $-2\pi$  (D)  $2\pi$

Ans. B

sol.  $I = \int_0^{2\pi} [\sin 2x(1 + \cos 3x)] dx \dots(i)$

$$\therefore \int_0^a f(x) dx = \int_0^a f(a-x) dx$$

$$\therefore I = \int_0^{2\pi} [-\sin 2x(1 + \cos 3x)] dx \quad \dots(ii)$$

By (i) + (ii)

$$2I = \int_0^{2\pi} (-1) dx$$

$$2I = -(x)_0^{2\pi}$$

$$\Rightarrow I = -\pi$$

15. The area bounded by the curve  $4y^2 = x^2(4-x)(x-2)$  is equal to :

वक्र  $4y^2 = x^2(4-x)(x-2)$  द्वारा परिबद्ध क्षेत्र का क्षेत्रफल है :

(A)  $\frac{3\pi}{8}$

(B)  $\frac{3\pi}{2}$

(C)  $\frac{\pi}{16}$

(D)  $\frac{\pi}{8}$

Ans. B

Sol.  $4y^2 = x^2(4-x)(x-2)$

$$y = \pm \frac{x}{2} \sqrt{(4-x)(x-2)}$$

$$(4-x)(x-2) \geq 0$$

$$(x-2)(x-4) \leq 0$$

$$x \in [2, 4]$$

$$y_1 = \frac{x}{2} \sqrt{(4-x)(x-2)}$$

$$y_2 = \frac{-x}{2} \sqrt{(4-x)(x-2)}$$

$$\text{Required area} = \int_2^4 (y_1 - y_2) dx$$

$$\text{Required area} = \int_2^4 x \sqrt{(4-x)(x-2)} dx = I$$

$$I = \int_2^4 (6-x) \sqrt{(x-2)(4-x)} \, dx$$

$$2I = \int_2^4 6 \sqrt{(x-2)(4-x)} \, dx$$

$$I = 3 \int_2^4 \sqrt{(x-2)(4-x)} \, dx$$

$$I = 3 \int_2^4 \sqrt{1-(x-3)^2} \, dx$$

$$I = 3 \times \frac{\pi}{2} = \frac{3\pi}{2}$$

16. The value of the integral  $\int_0^1 x \cot^{-1}(1-x^2+x^4) \, dx$  is :

समाकल  $\int_0^1 x \cot^{-1}(1-x^2+x^4) \, dx$  का मान है :

- (A)  $\frac{\pi}{4} - \log_e 2$       (B)  $\frac{\pi}{2} - \log_e 2$       (C)  $\frac{\pi}{2} - \frac{1}{2} \log_e 2$       (D)  $\frac{\pi}{4} - \frac{1}{2} \log_e 2$

Ans. D

**sol.**  $\int_0^1 x \cot^{-1}(1-x^2+x^4) \, dx = \int_0^1 x \tan^{-1}\left(\frac{1}{1+x^4-x^2}\right) \, dx$

$$\Rightarrow \int_0^1 x \tan^{-1}\left(\frac{x^2-(x^2-1)}{1+x^2(x^2-1)}\right) \, dx$$

$$\Rightarrow \int_0^1 x \tan^{-1} x^2 \, dx - \int_0^1 x \tan^{-1}(x^2-1) \, dx$$

Put  $x^2 = t \Rightarrow 2x \, dx = dt$ , (For 1<sup>st</sup> integration)

put  $x^2 - 1 = k \Rightarrow 2x \, dx = dk$  (For 2<sup>nd</sup> integration)

$$\Rightarrow \frac{1}{2} \int_0^1 x \tan^{-1} t \, dt - \frac{1}{2} \int_{-1}^0 1 \tan^{-1} k \, dk$$

$$\Rightarrow \frac{1}{2} \left( \int_0^1 t \tan^{-1} t \, dt - \int_0^1 \frac{t}{1+t^2} \, dt \right) - \frac{1}{2} \left( k \tan^{-1} k \int_{-1}^0 - \int_{-1}^0 \frac{k}{1+k^2} \, dk \right)$$

$$\Rightarrow \frac{1}{2} \left( \frac{\pi}{4} - \left( \frac{1}{2} \ln(1+t^2) \right) \int_0^1 \right) - \frac{1}{2} \left( 0 - \frac{\pi}{4} - \left( \frac{1}{2} \ln(1+k^2) \right) \int_{-1}^0 \right)$$

$$\Rightarrow \left( \frac{\pi}{8} - \frac{1}{4} \ln 2 \right) - \left( -\frac{\pi}{8} - \frac{1}{4} 10 - \ln 2 \right)$$

$$\Rightarrow \frac{\pi}{4} - \frac{1}{2} \ln 2$$

17. Let  $f$  be twice differentiable function defined on  $\mathbb{R}$  such that  $f(0) = 1$ ,  $f'(0) = 2$  and  $f''(x) \neq 0$  for all  $x \in \mathbb{R}$ . If

$$\begin{vmatrix} f(x) & f'(x) \\ f'(x) & f''(x) \end{vmatrix} = 0, \text{ for all } x \in \mathbb{R}, \text{ then value of } f(1) \text{ lies in the interval :}$$

माना  $\mathbb{R}$  पर एक फलन  $f$  दो बार अवकलनीय है, जिसके लिए  $f(0) = 1$ ,  $f'(0) = 2$  तथा  $f''(x) \neq 0$ ,  $x \in \mathbb{R}$  हैं। यदि

$$\begin{vmatrix} f(x) & f'(x) \\ f'(x) & f''(x) \end{vmatrix} = 0 \quad \forall x \in \mathbb{R} \text{ है, तो } f(1) \text{ का मान जिस अन्तराल में है, वह है :}$$

- (A) (3, 6)                      (B) (6, 9)                      (C) (0, 3)                      (D) (9, 12)

Ans. B

Sol.  $f(x) f''(x) - (f'(x))^2 = 0 \quad \forall x \in \mathbb{R}$

$$\frac{d}{dx} \left( \frac{f'(x)}{f(x)} \right) = 0$$

$$\frac{f'(x)}{f(x)} = c$$

$$\because f(0) = 1, f'(0) = 2 \Rightarrow c = 2$$

$$\frac{f'(x)}{f(x)} = 2$$

$$\int \frac{dy}{y} = \int 2dx$$

$$\ln y = 2x + \ln k \Rightarrow y = k e^{2x}$$

$$\because f(0) = 1 \Rightarrow k = 1$$

$$f(x) = e^{2x}$$

$$f(1) = e^2$$

18. Let  $f(x) = \int \frac{\sqrt{x}}{(1+x)^2} dx$  ( $x \geq 0$ ). then  $f(3) - f(1)$  is equal to

माना  $f(x) = \int \frac{\sqrt{x}}{(1+x)^2} dx$  ( $x \geq 0$ ) है, तो  $f(3) - f(1)$  बराबर है:

(A)  $\frac{\pi}{6} + \frac{1}{2} - \frac{\sqrt{3}}{4}$       (B)  $\frac{\pi}{12} + \frac{1}{2} - \frac{\sqrt{3}}{4}$       (C)  $-\frac{\pi}{6} + \frac{1}{2} + \frac{\sqrt{3}}{4}$       (D)  $-\frac{\pi}{12} + \frac{1}{2} + \frac{\sqrt{3}}{4}$

Ans. B

Sol. Let  $\sqrt{x} = \tan \theta$

$$x = \tan^2 \theta$$

$$\Rightarrow dx = 2 \tan \theta \sec^2 \theta d\theta$$

$$I = \int \frac{\tan \theta \cdot 2 \tan \theta \sec^2 \theta d\theta}{(1 + \tan^2 \theta)^2}$$

$$= \int \frac{2 \tan^2 \theta \sec^2 \theta}{\sec^4 \theta} d\theta$$

$$= \int \frac{2(\sec^2 \theta - 1)}{\sec^2 \theta} d\theta$$

$$= \int 2(1 - \cos^2 \theta) d\theta$$

$$= \int 2 \sin^2 \theta d\theta$$

$$= \int (1 - \cos 2\theta) d\theta$$

$$= \theta - \frac{\sin 2\theta}{2} + C$$

$$= \theta - \frac{\tan \theta}{1 + \tan^2 \theta} + C$$

$$f(x) = \tan^{-1}(\sqrt{x}) - \frac{\sqrt{x}}{1+x} + C$$

$$f(3) = \tan^{-1}(\sqrt{3}) - \frac{\sqrt{3}}{4} + C$$

$$f(1) = \tan^{-1}(1) - \frac{1}{2} + C$$

$$f(3) - f(1) = \tan^{-1}(\sqrt{3}) - \tan^{-1}(1) + \frac{1}{2} - \frac{\sqrt{3}}{4}$$

$$= \frac{\pi}{3} - \frac{\pi}{4} + \frac{1}{2} - \frac{\sqrt{3}}{4}$$

$$= \frac{\pi}{12} + \frac{1}{2} - \frac{\sqrt{3}}{4}$$

19. Let  $S(\alpha) = \{(x, y) : y^2 \leq x, 0 \leq x \leq \alpha\}$  and  $A(\alpha)$  is area of the region  $S(\alpha)$ .

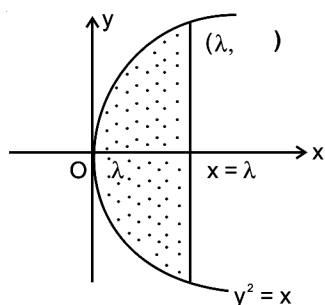
If for a  $\lambda$ ,  $0 < \lambda < 4$ ,  $A(\lambda) : A(4) = 2 : 5$ , then  $\lambda$  equals :

माना  $S(\alpha) = \{(x, y) : y^2 \leq x, 0 \leq x \leq \alpha\}$  तथा  $A(\alpha)$ , क्षेत्र  $S(\alpha)$  का क्षेत्रफल है। यदि किसी  $\lambda$ ,  $0 < \lambda < 4$  के लिए  $A(\lambda) : A(4) = 2 : 5$  है, तो  $\lambda$  बराबर है :

(A)  $2\left(\frac{2}{5}\right)^{1/3}$       (B)  $2\left(\frac{4}{25}\right)^{1/3}$       (C)  $4\left(\frac{2}{5}\right)^{1/3}$       (D)  $4\left(\frac{4}{25}\right)^{1/3}$

Ans. D

sol.



$$A(\lambda) = 2 \times \frac{2}{3} \lambda \times \sqrt{\lambda} = \frac{4}{3} \lambda$$

$$\Rightarrow \frac{A(\lambda)}{A(4)} = \frac{2}{5} \quad \frac{\lambda}{8} = \frac{2}{5}$$

$$\lambda = \frac{16^{2/3}}{5} = 4 \cdot \frac{4}{25}^{1/3}$$

20. Let  $\alpha \in (0, 1)$  and  $\beta = \log_e(1 - \alpha)$ . Let  $P_n(x) = x + \frac{x^2}{2} + \frac{x^3}{3} + \dots + \frac{x^n}{n}$ ,  $x \in (0, 1)$ . Then the integral

$\int_0^{\alpha} \frac{t^{50}}{1-t} dt$  is equal to

माना  $\alpha \in (0,1)$  तथा  $\beta = \log_e (1-\alpha)$  है। माना  $P_n(x) = x + \frac{x^2}{2} + \frac{x^3}{3} + \dots + \frac{x^n}{n}$ ,  $x \in (0,1)$  है। तो समाकलन  $\int_0^{\alpha} \frac{t^{50}}{1-t} dt$

बराबर है :

- (A)  $-(\beta + P_{50}(\alpha))$  (B)  $\beta + P_{50}(\alpha)$  (C)  $\beta - P_{50}(\alpha)$  (D)  $P_{50}(\alpha) - \beta$

**Ans.** A

**Sol.** 
$$\int_0^{\alpha} \frac{t^{50} - 1 + 1}{1-t} dt = -\int_0^{\alpha} (1 + t + \dots + t^{49}) dt + \int_0^{\alpha} \frac{1}{1-t} dt$$
$$= -\left( \frac{\alpha^{50}}{50} + \frac{\alpha^{49}}{49} + \dots + \frac{\alpha^1}{1} \right) + \left( \frac{\ln(1-t)}{-1} \right)_0^{\alpha}$$
$$= -P_{50}(\alpha) - \ln(1-\alpha)$$
$$= -P_{50}(\alpha) - \beta$$